

OSPF Routing and Configuration

5.1 Objectives

UnderStanding OSPF

Learn how To Config Routers With OSPF

Advertise directly connected networks using OSPF

..Verify OSPF configuration using different show commands

5.2 Equipment / Program Requirements:

eNSP

Cisco Packet Tracker

5.3 Experiment Setup:

The network topology consists of two routers connected together using an
.Ethernet link

.Each router is connected to a PC to form two different LANs

.Router 1 is connected to PC1 -

.Router 2 is connected to PC2 -

.The two routers are connected using a direct link -

:IP addressing is configured as follows

Network between routers: 11.0.0.0/30 -

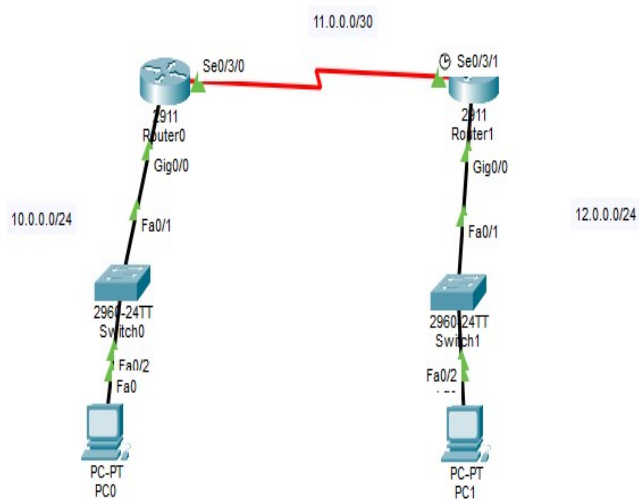
LAN 1: 10.0.0.0 /24 -

LAN 2: 12.0.0.0/24 -

OSPF is configured on both routers in Area 0 to enable dynamic routing and .allow communication between the two LANs

5.4 Simulation Results:

Cisco Packet Tracker Topology:



Router1 Configuration:

{router1

Physical Config CLI Attributes

IOS Command Line Interfac

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip add
Router(config-if)#ip address 12.0.0.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int se0/3/1
Router(config-if)#ip add
Router(config-if)#ip address 11.0.0.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 12.0.0.0 0.0.0.255
% Incomplete command.
Router(config-router)#network 12.0.0.0 0.0.0.255 area 0
Router(config-router)#network 11.0.0.0 0.0.0.3 area 0
Router(config-router)#do sh ip route
00:08:12: %OSPF-5-ADJCHG: Process 1, Nbr 11.0.0.1 on Serial0/3/1 from LOADING to FULL, Loading Done
```

Top

Router1 Route Table,OSPF, LSDB:

Router1

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 12.0.0.0 0.0.0.255
% Incomplete command.
Router(config-router)#network 12.0.0.0 0.0.0.255 area 0
Router(config-router)#network 11.0.0.0 0.0.0.3 area 0
Router(config-router)#do sh ip route
00:08:12: %OSPF-5-ADJCHG: Process 1, Nbr 11.0.0.1 on Serial0/3/1 from LOADING to FULL, Loading Done

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, Serial0/3/1
L       11.0.0.2/32 is directly connected, Serial0/3/1
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/24 is directly connected, GigabitEthernet0/0
L       12.0.0.1/32 is directly connected, GigabitEthernet0/0

Router(config-router)#do sh ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
11.0.0.1       0     FULL/-         00:00:38    11.0.0.1       Serial0/3/1
Router(config-router)#do sh ip ospf database
    OSPF Router with ID (12.0.0.1) (Process ID 1)

        Router Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum Link count
12.0.0.1       12.0.0.1      23          0x80000003    0x000f5d 3
11.0.0.1       11.0.0.1      23          0x80000003    0x001060 3
Router(config-router)#
Router#
```

Router 0 Configuration:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se0/3/0
Router(config-if)#ip add
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#ospf 1
^
% Invalid input detected at '^' marker.

Router(config)#ospf 1
^
% Invalid input detected at '^' marker.

Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 11.0.0.0 0.0.0.3 area 0
Router(config-router)#
00:08:21: %OSPF-5-ADJCHG: Process 1, Nbr 12.0.0.1 on Serial0/3/0 from LOADING to FULL, Los
Router(config-router)#
```

Router 0 Route Table , OSPF , LSDB:

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 11.0.0.0 0.0.0.3 area 0
Router(config-router)#
00:08:21: %OSPF-5-ADJCHG: Process 1, Nbr 12.0.0.1 on Serial0/3/0 from LOADING to FULL, Loading Done

Router(config-router)#do sh ip ospf database
      OSPF Router with ID (11.0.0.1) (Process ID 1)

      Router Link States (Area 0)

Link ID      ADV Router    Age          Seq#          Checksum Link count
11.0.0.1     11.0.0.1      208          0x80000003  0x001060 3
12.0.0.1     12.0.0.1      208          0x80000003  0x000f5d 3
Router(config-router)#do sh ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
12.0.0.1       0     FULL/-          00:00:32    11.0.0.2       Serial0/3/0

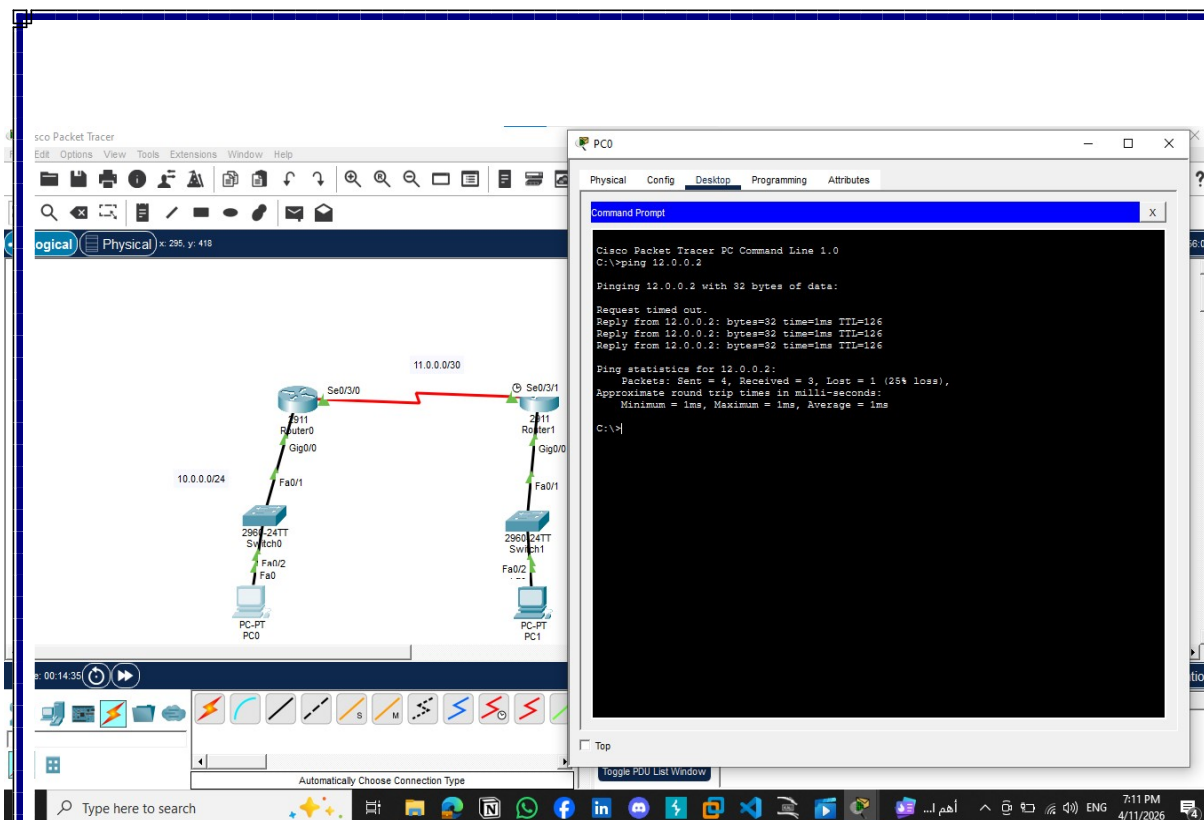
Router(config-router)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, Serial0/3/0
L       11.0.0.1/32 is directly connected, Serial0/3/0
    12.0.0.0/24 is subnetted, 1 subnets
O       12.0.0.0/24 [110/65] via 11.0.0.2, 00:03:46, Serial0/3/0

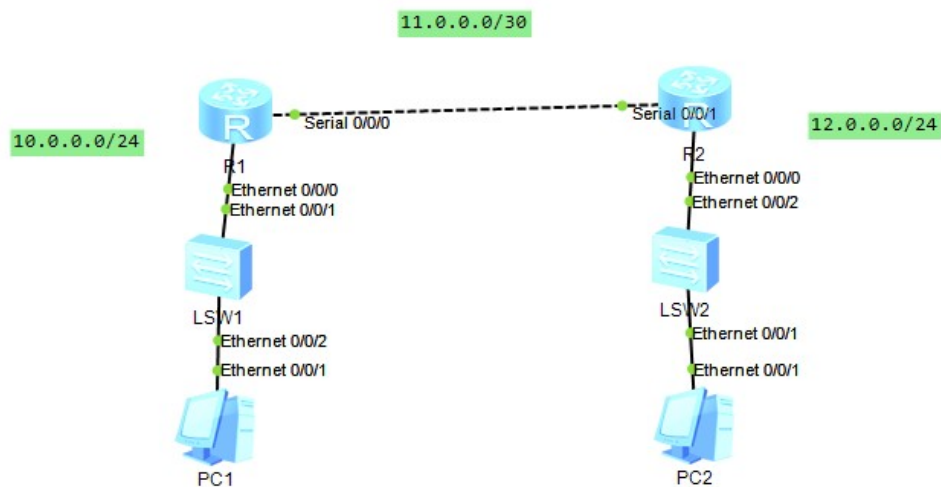
Router(config-router)#

```



eNSP:

Topology:



Configuration On R1:

```
R1
Error: Ambiguous command found at '^' position.
<Huawei>
<Huawei> u t l
Info: Current terminal logging is off.
<Huawei> u t m
Info: Current terminal monitor is off.
<Huawei> sys
Enter system view, return user view with Ctrl+Z.
[Huawei] int eth0/0/0
[Huawei-Ethernet0/0/0] ip add
[Huawei-Ethernet0/0/0] ip address 10.0.0.1 24
[Huawei-Ethernet0/0/0] int se0/0/0
[Huawei-Serial0/0/0] ip add
[Huawei-Serial0/0/0] ip address 11.0.0.1 30
[Huawei-Serial0/0/0] display ip rou
[Huawei-Serial0/0/0] display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 7      Routes : 7

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
10.0.0.0/24        Direct  0    0       D    10.0.0.1      Ethernet0/0/0
10.0.0.1/32        Direct  0    0       D    127.0.0.1     Ethernet0/0/0
11.0.0.0/30        Direct  0    0       D    11.0.0.1     Serial0/0/0
11.0.0.1/32        Direct  0    0       D    127.0.0.1     Serial0/0/0
11.0.0.2/32        Direct  0    0       D    11.0.0.2     Serial0/0/0
127.0.0.0/8        Direct  0    0       D    127.0.0.1     InLoopBack0
127.0.0.1/32       Direct  0    0       D    127.0.0.1     InLoopBack0

[Huawei-Serial0/0/0] q
[Huawei] ospf 1
[Huawei-ospf-1] area 0
[Huawei-ospf-1-area-0.0.0.0] network 24 10.0.0.0
^
Error: Wrong parameter found at '^' position.
[Huawei-ospf-1-area-0.0.0.0] network 10.0.0.0 24
^
Error: Wrong parameter found at '^' position.
[Huawei-ospf-1-area-0.0.0.0] network 10.0.0.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0] network 11.0.0.0 0.0.0.3
[Huawei-ospf-1-area-0.0.0.0] q
[Huawei-ospf-1] display ip rou
```

Configuration On R2

```
R2
The device is running!

<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]int se0/0/1
[Huawei-Serial0/0/1]ip add
[Huawei-Serial0/0/1]ip address 11.0.0.2 30
[Huawei-Serial0/0/1]int eth0/0/0
[Huawei-Ethernet0/0/0]ip add
[Huawei-Ethernet0/0/0]ip address 12.0.0.1 24
[Huawei-Ethernet0/0/0]display ip rou
[Huawei-Ethernet0/0/0]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 7      Routes : 7

Destination/Mask    Proto  Pre  Cost    Flags NextHop        Interface
-----
11.0.0.0/30        Direct  0    0        D 11.0.0.2    Serial0/0/1
11.0.0.1/32        Direct  0    0        D 11.0.0.1    Serial0/0/1
11.0.0.2/32        Direct  0    0        D 127.0.0.1    Serial0/0/1
12.0.0.0/24        Direct  0    0        D 12.0.0.1    Ethernet0/0/0
12.0.0.1/32        Direct  0    0        D 127.0.0.1    Ethernet0/0/0
127.0.0.0/8        Direct  0    0        D 127.0.0.1    InLoopBack0
127.0.0.1/32        Direct  0    0        D 127.0.0.1    InLoopBack0

[Huawei-Ethernet0/0/0]q
[Huawei]ospf 1
[Huawei-ospf-1]area 0
[Huawei-ospf-1-area-0.0.0.0]network 11.0.0.0 0.0.0.3
[Huawei-ospf-1-area-0.0.0.0]network 12.0.0.0 0.0.0.255
[Huawei-ospf-1-area-0.0.0.0]display ip rou
[Huawei-ospf-1-area-0.0.0.0]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 8      Routes : 8

Destination/Mask    Proto  Pre  Cost    Flags NextHop        Interface
-----
```

Routing Table For R1

```
[Huawei-ospf-1]display ip rou
[Huawei-ospf-1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 8      Routes : 8

Destination/Mask    Proto Pre  Cost   Flags NextHop         Interface
-----
10.0.0.0/24        Direct 0    0       D  10.0.0.1       Ethernet0/0/0
10.0.0.1/32        Direct 0    0       D  127.0.0.1       Ethernet0/0/0
11.0.0.0/30        Direct 0    0       D  11.0.0.1       Serial0/0/0
11.0.0.1/32        Direct 0    0       D  127.0.0.1       Serial0/0/0
11.0.0.2/32        Direct 0    0       D  11.0.0.2       Serial0/0/0
12.0.0.0/24        OSPF   10   1563     D  11.0.0.2       Serial0/0/0
127.0.0.0/8        Direct 0    0       D  127.0.0.1       InLoopBack0
127.0.0.1/32       Direct 0    0       D  127.0.0.1       InLoopBack0

[Huawei-ospf-1]display ospf peer

    OSPF Process 1 with Router ID 10.0.0.1
      Neighbors

Area 0.0.0.0 interface 11.0.0.1(Serial0/0/0)'s neighbors
Router ID: 11.0.0.2      Address: 11.0.0.2
State: Full Mode:Nbr is Master Priority: 1
DR: None BDR: None MTU: 0
Dead timer due in 29 sec
Retrans timer interval: 5
Neighbor is up for 00:01:07
Authentication Sequence: [ 0 ]

[Huawei-ospf-1]display ospf lsdb

    OSPF Process 1 with Router ID 10.0.0.1
      Link State Database

Area: 0.0.0.0
Type  LinkState ID  AdvRouter  Age  Len  Sequence  Metric
----  -
Router 10.0.0.1    10.0.0.1    71  60  80000004    1
Router 11.0.0.2    11.0.0.2    32  60  80000004   1562

[Huawei-ospf-1]
```

Routing Table For R2

```
R2
[Huawei-ospf-1-area-0.0.0.0]display ip rou
[Huawei-ospf-1-area-0.0.0.0]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 8          Routes : 8

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
10.0.0.0/24        OSPF    10   1563    D   11.0.0.1   Serial0/0/1
11.0.0.0/30        Direct  0     0       D   11.0.0.2   Serial0/0/1
11.0.0.1/32        Direct  0     0       D   11.0.0.1   Serial0/0/1
11.0.0.2/32        Direct  0     0       D   127.0.0.1   Serial0/0/1
12.0.0.0/24        Direct  0     0       D   12.0.0.1   Ethernet0/0/0
12.0.0.1/32        Direct  0     0       D   127.0.0.1   Ethernet0/0/0
127.0.0.0/8        Direct  0     0       D   127.0.0.1   InLoopBack0
127.0.0.1/32       Direct  0     0       D   127.0.0.1   InLoopBack0

[Huawei-ospf-1-area-0.0.0.0]display ospf peer

      OSPF Process 1 with Router ID 11.0.0.2
        Neighbors

Area 0.0.0.0 interface 11.0.0.2(Serial0/0/1)'s neighbors
Router ID: 10.0.0.1      Address: 11.0.0.1
State: Full  Mode:Nbr is Slave  Priority: 1
DR: None  BDR: None  MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00:00:23
Authentication Sequence: [ 0 ]

[Huawei-ospf-1-area-0.0.0.0]display ospf lsdb

      OSPF Process 1 with Router ID 11.0.0.2
        Link State Database

          Area: 0.0.0.0
Type  LinkState ID  AdvRouter   Age  Len  Sequence  Metric
-----
Router  10.0.0.1      10.0.0.1    44  60  80000004  1
Router  11.0.0.2      11.0.0.2    3  60  80000004  1562

[Huawei-ospf-1-area-0.0.0.0]
```

The screenshot shows the configuration window for PC2. It has tabs for Basic Config, Command, MCPacket, UdpPacket, and Console. The Basic Config tab is active, showing fields for Host Name, MAC Address (54-89-98-16-68-9D), and IPv4/IPv6 configurations. Under IPv4 Configuration, Static is selected, and fields are filled with IP Address: 12.0.0.2, Subnet Mask: 255.255.255.0, Gateway: 12.0.0.1, and DNS servers: 0.0.0.0. Under IPv6 Configuration, Static is selected, and fields are filled with IPv6 Address: ::, Prefix Length: 128, and IPv6 Gateway: ::. An Apply button is at the bottom right.

The screenshot shows the command console for PC1. It displays the output of a ping command to 12.0.0.2. The output shows 5 successful pings with varying round-trip times. The statistics at the bottom show 5 packets transmitted, 5 received, 0.00% loss, and a round-trip time of 94/118/156 ms.

```

Welcome to use PC Simulator!

PC>ping 120.0.0
host 120.0.0 unreachable

PC>ping 12.0.0.2

Ping 12.0.0.2: 32 data bytes, Press Ctrl_C to break
From 12.0.0.2: bytes=32 seq=1 ttl=126 time=156 ms
From 12.0.0.2: bytes=32 seq=2 ttl=126 time=110 ms
From 12.0.0.2: bytes=32 seq=3 ttl=126 time=94 ms
From 12.0.0.2: bytes=32 seq=4 ttl=126 time=125 ms
From 12.0.0.2: bytes=32 seq=5 ttl=126 time=109 ms

--- 12.0.0.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 94/118/156 ms

PC>

```

5.5 Conclusions:

The OSPF protocol allowed automatic route exchange between routers, which enabled successful communication between different LANs

..OSPF is more efficient than static routing in large and complex networks